

Development of a Two-Level, Four-Leg Smart Inverter for Microgrid Applications

Buck F. Brown, III

Department of Electrical and
Computer Engineering
Clemson University
North Charleston, SC, USA
buckb@clemson.edu

Jan Westman

Department of Electrical and
Computer Engineering
Clemson University
North Charleston, SC, USA
jwestma@clemson.edu

Johan Enslin

Department of Electrical and
Computer Engineering
Clemson University
North Charleston, SC, USA
jenslin@clemson.edu

Zheyu Zhang

Department of Electrical,
Computer, and Systems
Engineering
Rensselaer Polytechnic Institute
Troy, NY, USA
zhangz49@rpi.edu

Abstract—Microgrids are growing at a rapid pace, and as a result, power electronic inverters are more necessary than ever. However, the development of an adaptable inverter based on a changing control objective that also contains grid-related functions is lacking. This paper investigates the development of a novel two-level, four-leg smart inverter for a microgrid that can solve this problem. First, the inverter topology is discussed, modeled, and controlled using a standard dual-loop voltage control. Second, grid-forming functions are investigated. Third, an additional layout and control system are created for the grid-following inverter. Fourth, a simulation is constructed, and the capabilities of the smart inverter are highlighted.

Keywords—Four-wire smart inverter, inverter voltage control, inverter in microgrid, two-level four-leg smart inverter.

I. INTRODUCTION

As the utilization of renewable energy becomes more prominent, power electronic technologies become more important than ever. In particular, the implementation of power electronic converters in grid applications such as microgrids has drastically increased. According to the Department of Energy (DOE), a microgrid is a group of interconnected loads and distributed energy resources (DERs) within clearly defined electrical boundaries that acts as a single controllable entity with respect to the grid. A microgrid can connect and disconnect from the grid to enable it to operate in both grid-connected and islanded mode [1].

One of the most utilized types of converters in a microgrid is an inverter. Inverters are necessary to convert DC energy from photovoltaic (PV) panels to AC energy that can be used on the grid. In addition, inverters are necessary for battery energy storage systems (BESS), which store energy in the form of DC rather than the grid form of AC, and wind turbines. Often, microgrids implement what is known as a smart inverter—a traditional inverter combined with advanced controls, allowing for additional grid-related functions.

As part of a microgrid, an inverter can either be connected to the main grid (grid-connected mode) or it can be part of its own grid (islanded mode). While in either grid-connected or islanded mode, an inverter can operate as a grid-forming (GFM) or a grid-

following (GFL) inverter [2]. In GFM mode, an inverter is responsible for producing a set of output voltages. Usually, this set of output voltages is desired to be balanced. In GFL mode, an inverter is responsible for injecting current into its load, commonly the main grid. Therefore, an inverter always functions in one of four ways: as a GFM inverter connected to the grid, as a GFM inverter islanded, as a GFL inverter connected to the grid, and as a GFL inverter islanded. The two most common modes out of these four are GFM islanded and GFL grid connected.

A GFM inverter that is in an islanded system must provide a balanced set of output voltages and, ideally, provide virtual inertia. A GFL inverter that is connected to the grid must provide a controlled output current and contain some sort of P/f droop control to provide frequency support and Q/V droop control to provide voltage support.

These additional grid-related functions are often implemented in the control system of the smart inverter, and thus, are not heavily reliant on the topology of the smart inverter. However, the inherent control objective that comes with being a GFM inverter in an islanded system (providing a balanced set of output voltages) is topology dependent.

While a simple two-level, three-leg inverter can provide zero-sequence phase voltage, it cannot provide zero-sequence phase current because there is no path for such a current to flow. Therefore, if the three-phase load of the system is not balanced, there cannot be a balanced set of voltages at the load because of the resultant unbalanced load current. This issue can be resolved by adding a fourth wire to the system. Oftentimes, this wire is referred to as the neutral wire. The existence of this wire allows for zero-sequence current to flow in the circuit, and as a result, a balanced set of voltages can be produced regardless of load balancing. Therefore, a four-wire inverter is naturally a better choice when needing a balanced set of load voltages for any load condition. Additionally, four-wire inverters are capable of not only providing a balanced set of voltages at the output of the inverter but can also serve as active power filters (APFs), dynamic voltage restorers (DVRs), and unified power quality conditioners (UPQCs) [3].

This paper examines a novel two-level, four-leg smart inverter and its implementation in a microgrid. First, the inverter

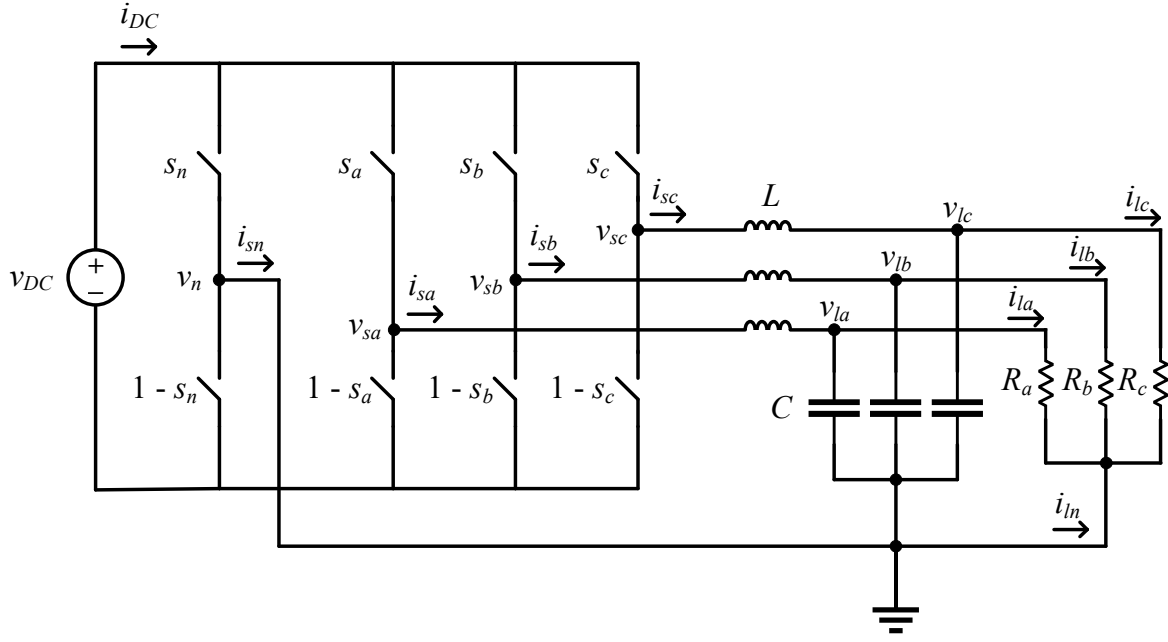


Fig. 1. Two-level, four-leg voltage source inverter.

topology and surrounding system are discussed. Second, the inverter is modeled mathematically, and a controller is designed. Third, specialized grid-related functions for both GFM inverters in an islanded system and GFL inverters in a grid-connected system are considered. Fourth, a simulation is built to showcase the effectiveness of the newly proposed inverter.

II. METHODOLOGY

A. Two-Level, Four-Leg Smart Inverter Topology

A standard two-level, four-leg voltage source inverter is shown in Fig. 1 [4]. This inverter has a two-level, four-leg topology, where the neutral line connects to the middle point of the fourth leg. The inverter is connected to a standard LC filter, and the output of the circuit is modeled as a simple resistive load.

B. System Topology

The inverter shown in Fig. 1 is connected in a system consisting of a PV array, a boost converter, and either an RL load for GFM modeling (as shown in Fig. 2(a)) or a grid emulating load for GFL modeling (as shown in Fig. 2(b)). The PV array serves as the sole source of generation. The boost converter interfaces between the PV array and the inverter to allow for a relatively constant DC input voltage. The boost converter is controlled through utilizing a maximum power point tracking (MPPT) scheme. This scheme uses a perturb and observe method to calculate the duty cycle of the boost converter. The boost converter in this system leverages two IGBTs to allow for bidirectional power flow if a BESS is desired to be implemented. The switches of the smart inverter are IGBTs with anti-parallel diodes.

C. Standard Modeling and Control

Before the smart inverter can be controlled to provide grid-related functions, it must first be controlled in standard ways.

TABLE I
SWITCH NETWORK

s_a	s_b	s_c	s_n	i_{DC}	v_{sa}	v_{sb}	v_{sc}
0	0	0	0	0	0	0	0
0	0	0	1	i_{sn}	$-v_{DC}$	$-v_{DC}$	$-v_{DC}$
0	0	1	0	i_{sc}	0	0	v_{DC}
0	0	1	1	$i_{sc} + i_{sn}$	$-v_{DC}$	$-v_{DC}$	0
0	1	0	0	i_{sb}	0	v_{DC}	0
0	1	0	1	$i_{sb} + i_{sn}$	$-v_{DC}$	0	$-v_{DC}$
0	1	1	0	$i_{sb} + i_{sc}$	0	v_{DC}	v_{DC}
0	1	1	1	$i_{sb} + i_{sc} + i_{sn}$	$-v_{DC}$	0	0
1	0	0	0	i_{sa}	v_{DC}	0	0
1	0	0	1	$i_{sa} + i_{sn}$	0	$-v_{DC}$	$-v_{DC}$
1	0	1	0	$i_{sa} + i_{sc}$	v_{DC}	0	v_{DC}
1	0	1	1	$i_{sa} + i_{sc} + i_{sn}$	0	$-v_{DC}$	0
1	1	0	0	$i_{sa} + i_{sb}$	v_{DC}	v_{DC}	0
1	1	0	1	$i_{sa} + i_{sb} + i_{sn}$	0	0	$-v_{DC}$
1	1	1	0	$i_{sa} + i_{sb} + i_{sc}$	v_{DC}	v_{DC}	v_{DC}
1	1	1	1	$i_{sa} + i_{sb} + i_{sc} + i_{sn} = 0$	0	0	0

To design a standard control system for the inverter, the inverter's transfer functions must be obtained.

1) *Switching Network Analysis*: The different switching states of the inverter are shown in Table I. Since there are four legs to this inverter, there are sixteen possible switching states. For each of these states, the input DC current and output AC voltages are written. By using Table I, the equations modeling the output voltages and input currents of the inverter can be derived. The results of this derivation are given as follows:

$$\vec{v}_{sph} = \vec{s}_{ph_n} \cdot v_{DC} \quad (1)$$

$$i_{DC} = \vec{s}_{ph_n}^T \vec{i}_{sph} \quad (2)$$

$$\vec{s}_{ph_n} = \begin{bmatrix} (s_a - s_n) \\ (s_b - s_n) \\ (s_c - s_n) \\ 0 \end{bmatrix} \quad (3)$$

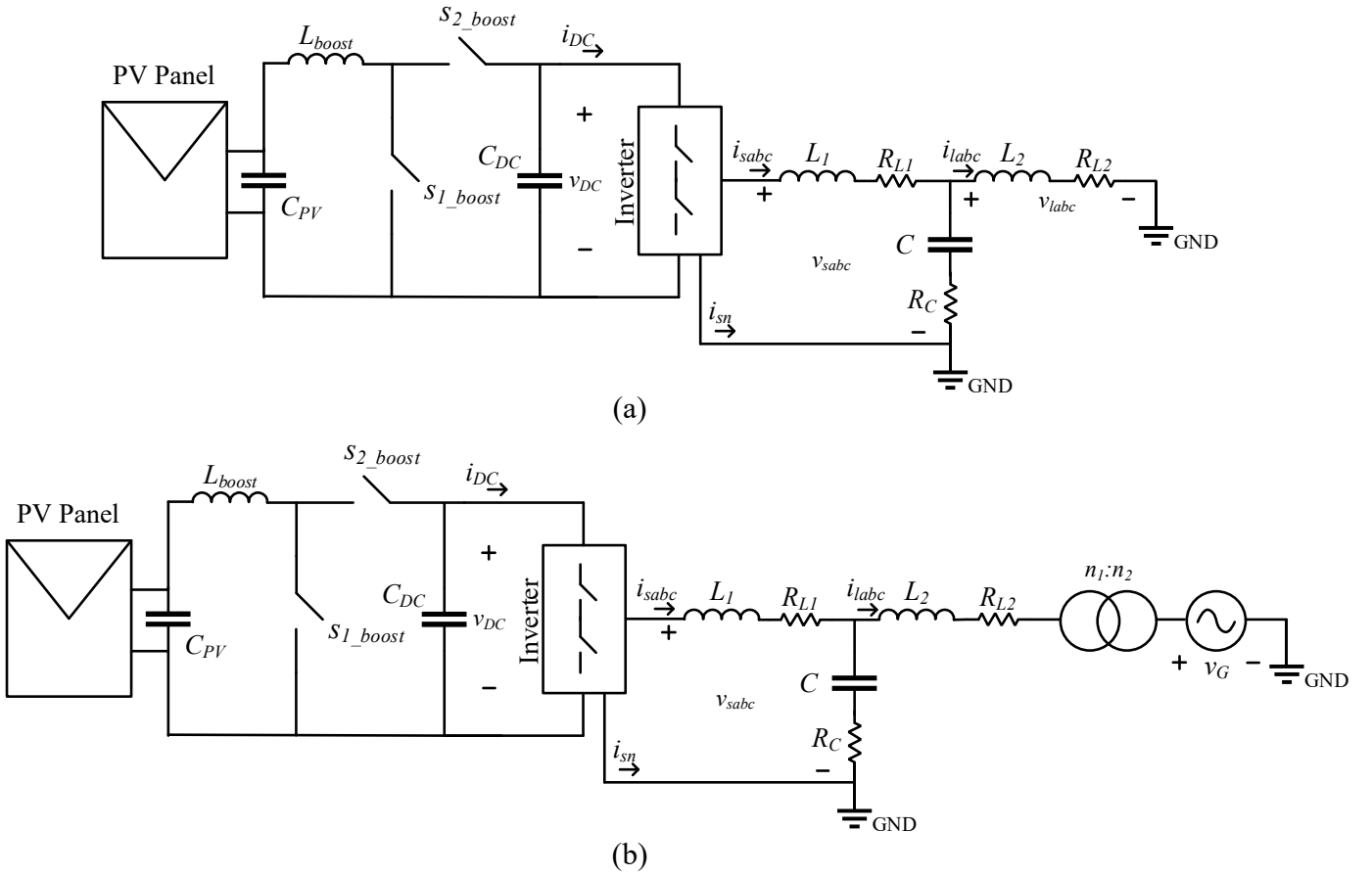


Fig. 2. Systems used to highlight the two-level, four-leg inverter's implementation in (a) GFM mode and (b) GFL mode.

2) *Switching, Average, and Small-signal Models*: The switching model of the system in abc coordinates is derived by using Kirchhoff's Voltage Law to write an equation for every inductor voltage in the circuit and by using Kirchhoff's Current Law to write an equation for every capacitor current in the circuit. After doing these two things, the equations modeling the inverter can be obtained. By averaging every variable, replacing the switching functions with their corresponding duty cycles, and applying a reference frame transformation, the switching model can be converted to an average model in $dq0$ coordinates. The small-signal model of the circuit is then found by replacing the average model variables with their small-signal counterparts. To eliminate any possible resonance due to the LC filter, a virtual resistance is added to this model. The equations that comprise the small-signal model are given below:

$$\frac{d\tilde{i}_{sdq0}}{dt} = \frac{V_{dc}\tilde{d}_{dq0} - \tilde{v}_{ldq0} - L \begin{bmatrix} 0 & -\omega & 0 \\ \omega & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \tilde{i}_{sdq0}}{L} \quad (4)$$

$$\frac{d\tilde{v}_{ldq0}}{dt} = \frac{\tilde{i}_{sdq0} - \tilde{i}_{ldq0} - \tilde{v}_{ldq0}/R - C \begin{bmatrix} 0 & -\omega & 0 \\ \omega & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \tilde{v}_{ldq0}}{C} \quad (5)$$

It should be noted that the DC input current is no longer a part of the modeling process since it is not the control objective. In addition, the input voltage source is ideal, so it has no small-signal component.

3) *Dual-loop Control*: Once the small-signal model of the inverter is derived, a state-space representation is obtained. From this, the transfer functions of the inverter are found. By using these transfer functions, a dual-loop feedback control system with inner loop current control and outer loop voltage control is used [5].

The design of the inner loop current control depends on the transfer functions relating the duty cycles to the source current. Through trial and error, it is determined that the virtual resistance, R , is to be 5Ω . After constructing the bode plots of the transfer functions, a PI compensator is selected based on a favorable existing phase margin of 106° . The current controller bandwidth, f_{ci} , is chosen to be a tenth of the switching frequency, which is initially chosen to be 16 kHz. The gain and zero location are chosen as follows:

$$k_{ii} = 10^{-\frac{44.4}{20}} = 0.006025 \quad (6)$$

$$\omega_{zi} = \frac{\omega_{ci}}{10} = 320\pi \text{ rad/s} \quad (7)$$

The outer loop voltage control depends on the transfer functions relating the source current to the load voltage. The

desired crossover frequency of the voltage controller, f_{cv} , is 160 Hz, which is one-tenth of the crossover frequency of the current controller, f_{ci} . However, since this frequency is very close to the cross-coupling frequency component of 120 Hz, it is desired to double f_{cv} , making it one-fifth of f_{ci} . Since the phase margin at this frequency is already sufficient, only a PI compensator is necessary. The parameters of the PI compensator are calculated as follows:

$$k_{iv} = 10^{-\frac{19.5}{20}} = 0.10593 \quad (8)$$

$$\omega_{zv} = \frac{\omega_{cv}}{10} = 201.06 \frac{\text{rad}}{\text{s}} \quad (9)$$

It is well understood that unbalanced loads result in negative-sequence currents. These are AC components of the line currents that rotate in the negative direction at the fundamental frequency. In the synchronously rotating reference frame, these components appear to have double the line frequency in dq coordinates and line frequency in θ coordinates. The design of the outer voltage loop bandwidth constrains the voltage regulator open loop gain, preventing it from being sufficiently large at double the fundamental frequency to effectively eliminate these components. Therefore, parallel proportional resonant (PR) compensation is added to the voltage control. This is accomplished by first applying a notch filter to eliminate the non-negative sequence components of the $dq\theta$ voltage error. The notch filter transfer function is given as follows:

$$H_{nf} = \frac{s^2 + \omega_r^2}{s^2 + k\omega_r s + \omega_r^2} \quad (10)$$

A gain, k_{pr} , is also added in the forward loop path to decrease the negative sequence component steady-state error.

In addition to the closed-loop feedback scheme discussed previously, a feedforward scheme is implemented to try to eliminate cross-coupling variables and disturbances.

D. 3D SVM

The two most common types of modulation are sinusoidal pulse width modulation (SPWM) and space vector modulation (SVM). In applications where the output voltage is the control objective, SVM is used due to its 15% voltage advantage over SPWM. Traditional SVM relies on the conversion of abc variables to $\alpha\beta$ variables. However, since the inverter in this paper contains a fourth wire, meaning negative and zero-sequence components may exist, abc variables can no longer be converted to just $\alpha\beta$ variables; they must be converted to $\alpha\beta\gamma$ variables. Thus, traditional SVM cannot be used, and a 3D SVM scheme is created utilizing the work done in [4] and [6].

E. GFM Functions

The main function of the dual-loop control previously derived is to maintain a balanced set of output voltages. As a result, when the inverter is operating in GFM mode, it is assumed to be in an islanded system. Therefore, it is treated as the main source of AC voltage in the system rather than a secondary source.

F. GFL Functions

Most inverters in GFL mode are assumed to operate in a system connected to the grid. Thus, this is the operating

condition considered here. As stated previously, the goal of the dual-loop control is to maintain a balanced set of output voltages. However, when an inverter is in GFL mode, the control objective changes from providing voltage to providing current. In addition, since the inverter is assumed to be connected to the grid, all grid-related functions of the smart inverter involve some sort of current injection into the grid.

In GFL mode, an LC filter is not acceptable because the capacitor is in parallel with the grid (an ideal voltage source), so an L or LCL filter is required. As a result, the inner loop current control previously derived is no longer applicable. Because of this obstacle, a new system is utilized to show how the smart inverter can operate when in GFL mode [7]. The system given in [7] contains the following: P/f droop control, standard current control, and Q/V droop control.

1) *P/f Droop Control*: The purpose of P/f droop control is to generate or absorb active power when the frequency of the grid varies from 60 Hz. To accomplish this task, P/f droop control becomes part of the boost converter controller [8]. This scheme reserves active power such that the boost converter operates at conditions lower than the maximum power point. Therefore, when frequency regulation is needed, active power is accessible. The equation modeling the P/f droop control is given below:

$$P_{pv}^* = \eta P_{MPPE} + D_p (f_n - f_{grid}) \quad (11)$$

where η determines reserved active power, D_p is the P/f droop constant determined by IEEE Std. 1547, f_n is the rated grid frequency, and f_{grid} is the grid frequency obtained by a PLL.

2) *Standard Current Control*: The current controller enables inverter output currents to track their references. The d -axis current reference is provided by the DC voltage control, and the q -axis current reference is provided by the Q/V control. An additional feedforward scheme is used to eliminate the cross-coupling between the d and q -axis variables. The current controller outputs the d and q -axis duty cycles of the inverter, which are converted to PWM signals. In addition, the fourth leg of the inverter is constantly controlled to be off since zero and negative-sequence currents are not desired. Investigation of the advantages of utilizing the fourth leg of the inverter for GFL applications is left for future study.

3) *Q/V Droop Control*: The purpose of Q/V droop control is to vary the q -axis current when the voltage of the grid varies from its nominal value. As a result of this current variation, the reactive power, Q , of the system is varied, and hence, the voltage of the grid is regulated. Associated with the Q/V droop controller is the droop constant, D_q , determined by IEEE Std. 1547.

III. RESULTS

A. Simulation Overview

Two simulations are constructed using MATLAB/Simulink: one for the smart inverter in GFM mode operating in an islanded system (demonstrated in Fig. 2(a)), and one for the

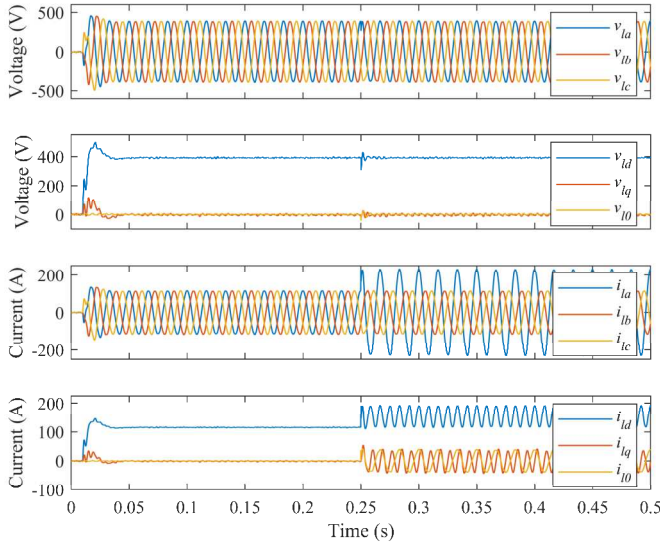


Fig. 3. Load voltage and current under unbalanced resistive load change.

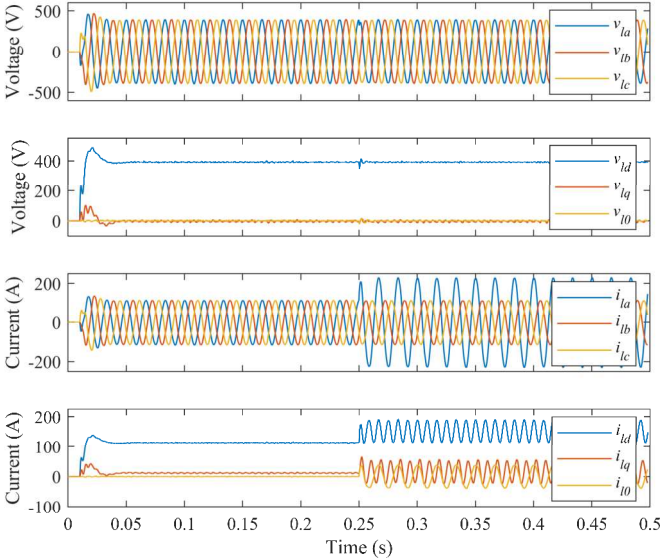


Fig. 4. Load voltage and current under unbalanced impedance load change.

smart inverter in GFL mode operating in a grid-connected system (demonstrated in Fig. 2(b)). Additionally, many electrical toolboxes such as PLLs, abc to $dq0$ transformations, IGBT modules, measurement tools, etc., are utilized.

B. Simulation of GFM Inverter

The first simulation is designed to test the system when a balanced resistive load becomes unbalanced. The initial load of each phase of the circuit is a 3.4Ω resistor. Therefore, the system is initially balanced. At 0.25 s, a 3.4Ω resistor is switched in parallel with the original resistor in phase a only. As a result, phases b and c have 3.4Ω at their load, whereas phase a has 1.7Ω at its load. Therefore, the system is greatly unbalanced. The results of this test are shown in Fig. 3. As can be seen in the figure, the load voltage is maintained at its desired value. In addition, the abc load currents become unbalanced, as expected. Therefore, a greater ripple is seen in the d , q , and 0 -

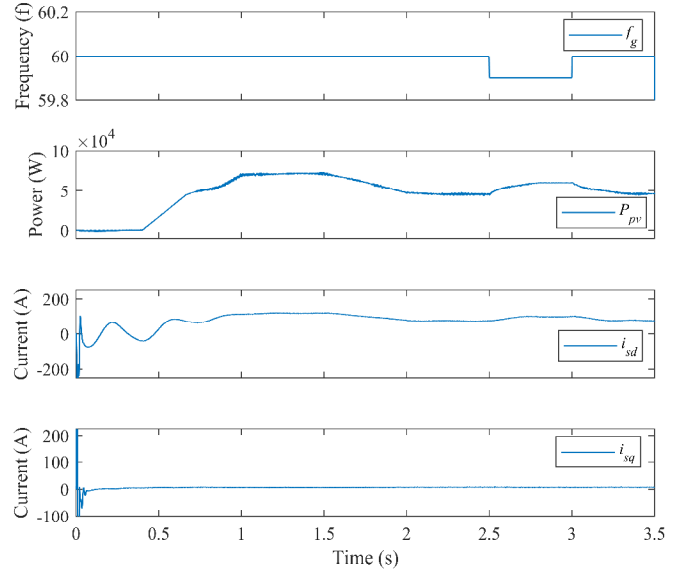


Fig. 5. Variation in grid frequency to illustrate P/f droop control.

axis load currents due to the existence of the negative-sequence load current that naturally comes with an unbalanced system.

The second simulation is designed to test the system when a balanced RL load becomes unbalanced. The initial load of each phase of the circuit is a 3.4Ω resistor in series with a 1 mH inductor. Therefore, the system is initially balanced. At 0.25 s, a 3.4Ω resistor and a 1 mH inductor are switched in parallel with the original impedance in phase a only. Therefore, the system is now greatly unbalanced. The results of this test are shown in Fig. 4. As can be seen in the figure, even with the unbalanced nature of the load, the load voltage is maintained at its desired value. In addition, the abc load currents become unbalanced, as expected. Therefore, a greater ripple is seen in the d , q , and 0 -axis load currents due to the existence of the negative-sequence load current that naturally comes with an unbalanced system. The main difference between this test and the previous test is the existence of the q -axis current throughout operation due to the inductor.

From these two tests, it is concluded that the control system is successful at maintaining a constant output voltage even if the load of the inverter is changed.

C. Simulation of GFL Inverter

According to IEEE Std. 1547-2018, the grid frequency should be regulated to be between 58.8 Hz and 61.2 Hz when connected to a microgrid. Therefore, it is the smart inverter's job to regulate this frequency. The test shown in Fig. 5 illustrates the effectiveness of the P/f droop control in this system. Before 2.5 s, it is seen that the system follows the active power reference as the irradiance is varied. At 2.5 s, the frequency of the grid is decreased to 59.9 Hz. Right after 2.5 s, the active power of the PV array is controlled to increase to compensate for this frequency dip. At 3 s, the grid frequency is returned to 60 Hz, and the active power of the PV array once again follows the active power reference. Therefore, the smart inverter in GFL mode can provide frequency support.

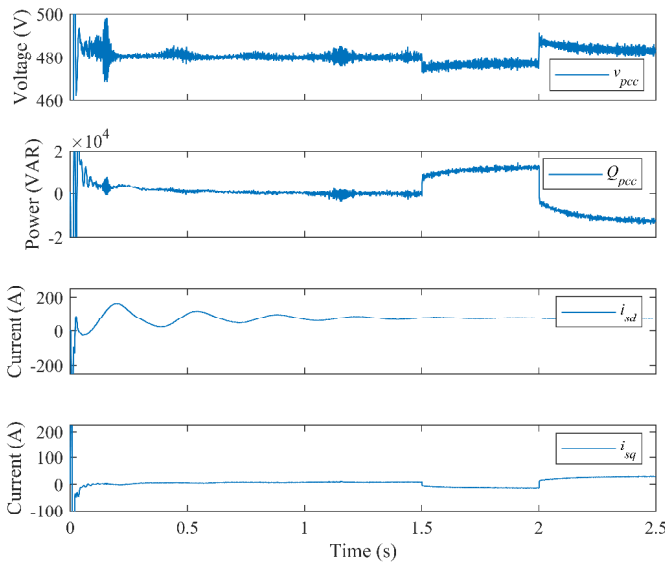


Fig. 6. Variation in grid voltage to illustrate Q/V droop control.

According to IEEE Std. 1547-2018, the grid voltage should stay between 0.95 p.u. and 1.05 p.u. The Q/V control does this by increasing the reactive power of the system if the voltage decreases and by decreasing the reactive power of the system if the voltage increases. The reactive power of the system is controlled by varying the q -axis current of the inverter. The test shown in Fig. 6 illustrates the effectiveness of the Q/V droop control in this system. Before 1.5 s, the system operates as expected. At 1.5 s, the voltage of the grid is decreased to 0.98 p.u. In response, the reactive power of the system increases to counterbalance the voltage. At 2.0 s, the voltage of the grid is increased to 1.02 p.u. In response, the reactive power of the system decreases to counterbalance the voltage. Therefore, the smart inverter in GFL mode can provide voltage support.

IV. CONCLUSION

In conclusion, this paper proposes a novel two-level, four-leg smart inverter and examines it in a simple microgrid layout. The inverter topology is selected, modeled, and controlled using dual and closed-loop voltage control. Then, additional grid-related functions of the smart inverter are developed. The smart inverter is placed in a microgrid and is tested both in islanded and grid-connected modes. Simulated results show that the smart inverter designed in this paper can produce desired voltage and current outputs, as well as provide additional grid-related functions such as P/f droop control and Q/V droop control. The following are some topics that could be considered for future studies: GFM inverters tied to the grid with an LCL filter, the advantages of a fourth leg in GFL inverters, and the benefits of grid-related functions.

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